



NORTH SOUTH UNIVERSITY
Examination Book
Department of Mathematics and Physics

ASSIGNMENT 02**SEMESTER: SUMMER 2026****General Physics II****Faculty: RUH****Course Code: PHY 108****Total Points: 20****STUDENT'S INFORMATION**Section :

Course Teacher's Initial : RUH

Student ID :

Course Code : ...PHY-108 ...

Full Name : Department :

Student's Signature with Date :

Last Date of Submission: 17 APRIL 2026**INSTRUCTIONS:**

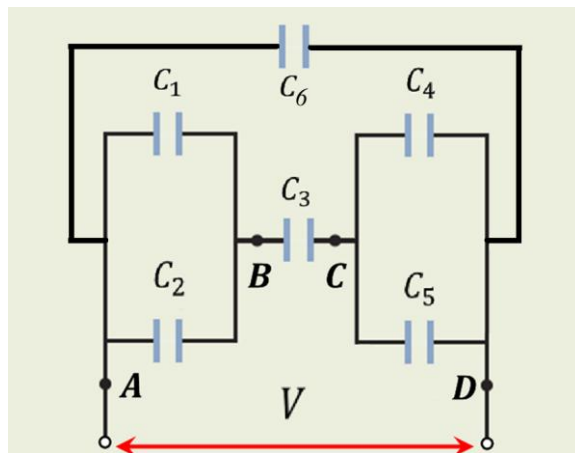
- ANSWER ALL QUESTIONS.
- WRITE CLEARLY WITH A PEN.
- SHOW STEP-BY-STEP CALCULATION.
- LATE SUBMISSIONS WILL NOT BE ACCEPTED.
- SUBMIT YOUR OWN HANDWRITTEN PAPER/HARD COPY DURING THE CLASS.
- DO NOT COPY FROM OTHERS. THERE WILL BE A MARK DEDUCTION FOR COPYING.

QUESTION 01

- (A) (i) What is the expression for the capacitance of an empty parallel-plate capacitor with metal plates that each have an area of A , separated by a distance d between them?
- (ii) What type of energy is stored by the capacitor?
- (iii) What will happen if substantially high voltage is applied between the plates of the capacitor?
- (iv) There are three ways to increase the capacitance of a parallel-plate capacitor. What are those?

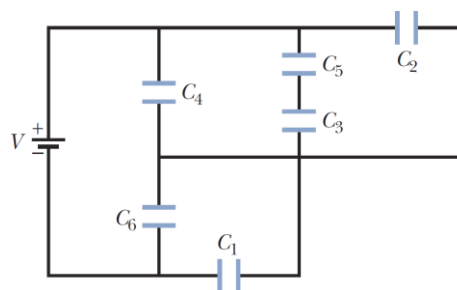
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- (B) Find the equivalent capacitance for the combination of capacitances shown in Fig, across which potential difference V is applied. Assume $C_1 = 12.0 \mu F$, $C_2 = 5.30 \mu F$, $C_3 = 4.50 \mu F$, $C_4 = 17.0 \mu F$, $C_5 = 9.30 \mu F$, and $C_6 = 10.0 \mu F$.



1

- (C) In Figure, a $20.0 V$ battery is connected across capacitors of capacitances $C_1 = C_6 = 3.00 \mu F$ and $C_3 = C_5 = 2.00 \mu F$. $C_2 = 2.00 \mu F$, $C_4 = 4.00 \mu F$. What are (i) the equivalent capacitance C_{eq} of the capacitors and the charge stored by C_{eq} ? What are (ii) V_1 and q_1 of capacitor 1, (iii) V_2 and q_2 of capacitor 2, and (iv) V_3 and q_3 of capacitor 3?

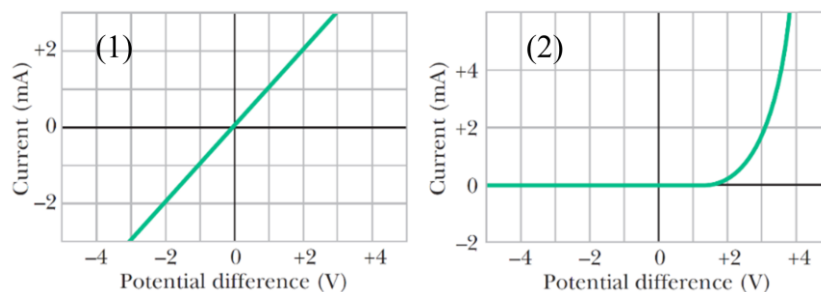


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QUESTION 02

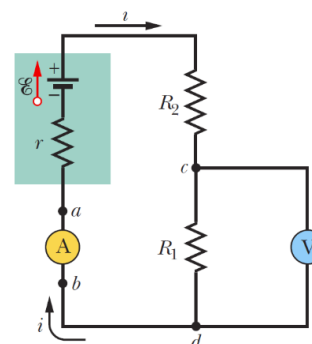
- (A) (i) What is the direction of the electric current? 1
- (ii) What is the primary difference between an ideal emf device and a real emf device?
- (iii) Let us consider a cylindrical having radius of r and length of L . Whether the resistivity of the conductive cylinder will increase or decrease if only the radius will be reduced to $\frac{r}{2}$ remaining other things constant.

- (B) (i) Write down the expression of current-voltage relationship. Figure below shows the I-V plot of resistor and diode. Why they exhibit different characteristics? Explain. 1



- (ii) Draw the schematic of simple RC circuit? What defines the behavior of the RC circuit, considering a voltage source is connected to an RC circuit.

- (C) (i) In Figure, assume that $\varepsilon_{EMF} = 3.0 \text{ V}$, $r = 100 \Omega$, $R_1 = 250 \Omega$, and $R_2 = 300 \Omega$. If the voltmeter resistance R_V is $5.0 \text{ k}\Omega$, what percent error does it introduce into the measurement of the potential difference across R_1 ? Ignore the presence of the ammeter.



- (ii) A $15.0 \text{ k}\Omega$ resistor and a capacitor are connected in series, and then a 12.0 V potential difference is suddenly applied across them. The potential difference across the capacitor rises to 5.00 V in $1.30 \mu\text{s}$. Calculate the time constant of the circuit. And find the capacitance of the capacitor.

QUESTION 03

- (A) (i) What is the direction of the magnetic field for a bar magnet? Write down the properties of the magnetic field lines. 1

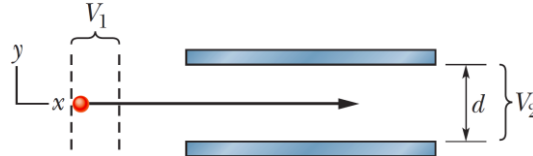
(ii) What is magnetic force? How to find the direction of the magnetic force knowing the direction of the current and magnetic field?

(iii) What is Hall effect?

- (B) (i) Draw the magnetic field of a bar magnet with north and south pole. 1

(ii) Draw the magnetic field of a solenoid.

- (C) (i) In Figure, an electron accelerated from rest through potential difference $V_1 = 1.00 \text{ kV}$ enters the gap between two parallel plates having separation $d = 20.0 \text{ mm}$ and potential difference $V_2 = 100 \text{ V}$. 2



The lower plate is at the lower potential. Neglect fringing and assume that the electron's velocity vector is perpendicular to the electric field vector between the plates. In unit-vector notation, what uniform magnetic field allows the electron to travel in a straight line in the gap?

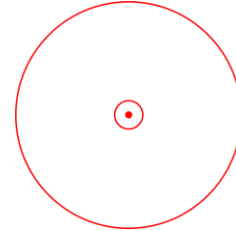
(ii) An electron moves in a circle of radius $r = 5.29 \times 10^{-11} \text{ m}$ with speed $2.19 \times 10^6 \text{ m/s}$. Treat the circular path as a current loop with a constant current equal to the ratio of the electron's charge magnitude to the period of the motion. If the circle lies in a uniform magnetic field of magnitude $B = 7.10 \text{ mT}$, what is the maximum possible magnitude of the torque produced on the loop by the field?

QUESTION 04

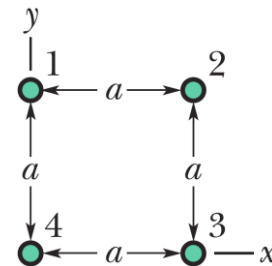
- (A) (i) Write down ampere's law? 1
 (ii) What is Amperian loop?
 (iii) How to calculate the net current for ampere's law?

- (B) (i) Draw the magnetic field line around a current carrying straight wire lying vertical and current is flowing from bottom to top direction. 1

(ii) A conducting wire loop is placed in a magnetic field that is coming out of the surface of this loop and is growing, as illustrated in the Figure. Draw the arrow to show the direction of the induced current (i_{ind}) that should be generated in the loop.



- (C) (i) In Figure, four long straight wires are perpendicular to the page, and their cross sections form a square of edge length $a = 8.50\text{ cm}$. Each wire carries 15.0 A , and all the currents are out of the page. In unit-vector notation, what is the net magnetic force *per meter of wire length* on wire 1? 2



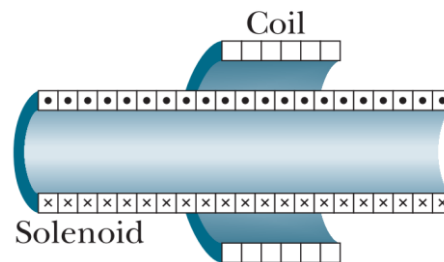
- (ii) A toroid having a square cross section, 5.00 cm on a side, and an inner radius of 15.0 cm has 500 turns and carries a current of 0.800 A . What is the magnetic field inside the toroid at the inner radius and the outer radius?

QUESTION 05

- (A) (i) What causes the formation of induced emf in a conducting loop? If two coils are placed close to each other. Name the factors that dictate the mutual inductance between the pair of coils. 1
- (ii) Write down Faraday's law. Explain magnetic flux.
- (iii) Write Lenz's law? Explain induced current.
- (iv) Explain Eddy current.
- (v) Explain self-induction.

- (B) (i) Draw the schematic of simple RL circuit? 1
- (ii) What defines the behavior of the RL circuit, considering a voltage source is connected to an LR circuit

- (C) (i) In Fig. a 120 – turn coil of radius 1.8 cm and resistance 5.3 Ω is coaxial with a solenoid of 220 turns/cm and diameter 3.2 cm. The solenoid current drops from 1.5 A to zero in time interval $\Delta t = 25$ ms. What current is induced in the coil during Δt ? 2



- (ii) Figure *a* shows a circuit consisting of an ideal battery with emf $\mathcal{E} = 6.00 \mu V$, a resistance R , and a small wire loop of area 5.0 cm^2 . For the time interval $t = 10 \text{ s}$ to $t = 20 \text{ s}$, an external magnetic field is set up throughout the loop. The field is uniform, its direction is into the page in Fig. *a*, and the field magnitude is given by $B = at$, where B is in teslas, a is a constant, and t is in seconds. Figure *b* gives the current i in the circuit before, during, and after the external field is set up. The vertical axis scale is set by $i_s = 2.0 \text{ mA}$. Find the constant a in the equation for the field magnitude.

